

# Foundations of Machine translation

[Machine translation]

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## 1.0 Content Block

In information extraction, named entities, in a narrow sense, refer to concrete or abstract entities in the real world such as people, organizations, companies, and places that have a proper name: George Washington, Chicago, Microsoft. It also refers to expressions of time, space and quantity such as 1 July 2011, \$500. In the sentence "Smith is the president of Fabrionix" both Smith and Fabrionix are named entities, and can be further qualified via first name or other information; "president" is not, since Smith could have earlier held another position at Fabrionix, e.g. Vice President. The term rigid designator is what defines these usages for analysis in statistical machine translation. Named entities must first be identified in the text; if not, they may be erroneously translated as common nouns, which would most likely not affect the BLEU rating of the translation but would change the text's human readability. They may be omitted from the output translation, which would also have implications for the text's readability and message. Transliteration includes finding the letters in the target language that most closely correspond to the name in the source language. This, however, has been cited as sometimes worsening the quality of translation. For "Southern California" the first word should be translated directly, while the second word should be transliterated. Machines often transliterate both because they treated them as one entity. Words like these are hard for machine translators, even those with a transliteration component, to process. Use of a "do-not-translate" list, which has the same end goal – transliteration as opposed to translation – still relies on correct identification of named entities. A third approach is a class-based model. Named entities are replaced with a token to represent their "class"; "Ted" and "Erica" would both be replaced with "person" class token. Then the statistical distribution and use of person names, in general, can be analyzed instead of looking at the distributions of "Ted" and "Erica" individually, so that the probability of a given name in a specific language will not affect the assigned probability of a translation. A study by Stanford on improving this area of translation gives the examples that different probabilities will be assigned to "David is going for a walk" and "Ankit is going for a walk" for English as a target language due to the different number of occurrences for each name in the training data. A frustrating outcome of the same study by Stanford (and other attempts to improve named recognition translation) is that many times, a decrease in the BLEU scores for translation will result from the inclusion of methods for named entity translation.

## 2.0 Content Block

**Non-standard speech** One of the major pitfalls of MT is its inability to translate non-standard language with the same accuracy as standard language. Heuristic or statistical based MT takes input from various sources in standard form of a language. Rule-based translation, by nature, does not include common non-standard usages. This causes errors in translation from a vernacular source or into colloquial language. Limitations on translation from casual speech present issues in the use of machine translation in mobile devices.

## 3.0 Content Block

There are many factors that affect how machine translation systems are evaluated. These factors include the intended use of the translation, the nature of the machine translation software, and the nature of the translation process. Different programs may work well for different purposes. For example, statistical machine translation (SMT) typically outperforms example-based machine translation (EBMT), but researchers found that when evaluating English to French translation, EBMT performs better. The same concept applies for technical documents, which can be more easily translated by SMT because of their formal language. In certain applications, however, e.g., product descriptions written in a controlled

language, a dictionary-based machine-translation system has produced satisfactory translations that require no human intervention save for quality inspection. There are various means for evaluating the output quality of machine translation systems. The oldest is the use of human judges to assess a translation's quality. Even though human evaluation is time-consuming, it is still the most reliable method to compare different systems such as rule-based and statistical systems. Automated means of evaluation include BLEU, NIST, METEOR, and LEPOR. Relying exclusively on unedited machine translation ignores the fact that communication in human language is context-embedded and that it takes a person to comprehend the context of the original text with a reasonable degree of probability. It is certainly true that even purely human-generated translations are prone to error. Therefore, to ensure that a machine-generated translation will be useful to a human being and that publishable-quality translation is achieved, such translations must be reviewed and edited by a human. The late Claude Piron wrote that machine translation, at its best, automates the easier part of a translator's job; the harder and more time-consuming part usually involves doing extensive research to resolve ambiguities in the source text, which the grammatical and lexical exigencies of the target language require to be resolved. Such research is a necessary prelude to the pre-editing necessary in order to provide input for machine-translation software such that the output will not be meaningless. In addition to disambiguation problems, decreased accuracy can occur due to varying levels of training data for machine translating programs. Both example-based and statistical machine translation rely on a vast array of real example sentences as a base for translation, and when too many or too few sentences are analyzed accuracy is jeopardized. Researchers found that when a program is trained on 203,529 sentence pairings, accuracy actually decreases. The optimal level of training data seems to be just over 100,000 sentences, possibly because as training data increases, the number of possible sentences increases, making it harder to find an exact translation match. Flaws in machine translation have been noted for their entertainment value. Two videos uploaded to YouTube in April 2017 involve two Japanese hiragana characters ■■ (e and gu) being repeatedly pasted into Google Translate, with the resulting translations quickly degrading into nonsensical phrases such as "DECEARING EGG" and "Deep-sea squeeze trees", which are then read in increasingly absurd voices; the full-length version of the video currently has 7.1 million views as of August 2025.